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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCSC100 - DATA VISUALIZATION USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Scheme Of Evaluation

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Arrange the following steps in the correct order to create a Matplotlib plot: (Customize plot, Show plot, Import library, Create figure, Add data). [K] / [CO1]

   Ans:**Import library- Create figure- Add data- Customize plot-Show plot. (Give 2 marks accordingly)**

2. Name two Python libraries besides Matplotlib used for data visualization. [K] / [CO1]

   Ans: Write any 2 libraries such as **Seaborn, Plotly, Bokeh**, **Altair**, **ggplot, Pygal**, **Geopandas / Folium (1 mark each)**

3. Explain the term ‘hover’. [U] / [CO2]

   Definition - 2 marks

4. Write down 2 best practices for data visualization. [U] / [CO2]

   Any 2 best practices (1 mark each)

5. Recall the role of dcc.Graph() in Dash. [K] / [CO2]

   Ans: **To render interactive plots and charts in a web application. (2marks)**

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Describe the importance of data visualization in decision-making. [K] / [CO1]

   Describe any 3 importance – 5 marks

7. Discuss the advantages and disadvantages of using pie charts for data representation. [U] / [CO2]

   Advantages – 2.5 marks

   Disadvantages – 2.5 marks

8. Discuss when to use box plots and violin plots for statistical visualization. [U] / [CO1]

   Ans: **Describe box plots and violin plots (1.5 marks each)**

   Examples (1 mark each)

   Both are used to visualize the distribution of a dataset.

9. Briefly explain some interactive function available. [U] / [CO2]

   Explanation of any three interactive functions – 5 marks

10. Discuss about Plotly Graph Objects. [U] / [CO2]

    Explanation – 5 marks

11. What is a multi-plot grid in Seaborn? How is it useful? [K] / [CO1]

    Definition – 3 marks

    Use – 2 marks

12. Describe how to use a built-in dataset in Plotly Express for visualization with an example. [U] / [CO2]

    Description – 3 marks

    Example – 2 marks

[5x5 = 25]